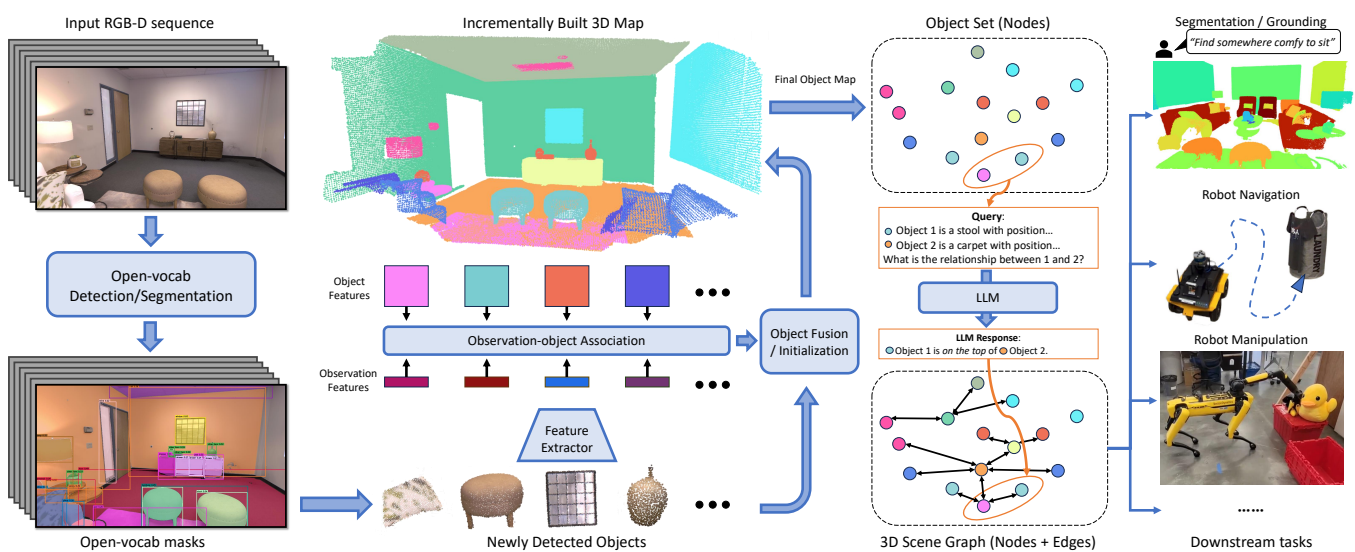

Input RGB-D sequence
Open-vocab Detection/Segmentation
Open-vocab masks
Incrementally Built 3D Map
Final Object Map
Object Features
Observation-object Association
Observation Features
Feature Extractor
Newly Detected Objects
Object Fusion / Initialization
Object Set (Nodes)
Query:
Object 1 is a stool with position...
Object 2 is a carpet with position...
What is the relationship between 1 and 2?
LLM
LLM Response:
Object 1 is on the top of Object 2.
3D Scene Graph (Nodes + Edges)
Segmentation / Grounding
"Find somewhere comfy to sit"
Robot Navigation
Robot Manipulation
.....
Downstream tasks